

EP4DIS MSc Dissertation (Research Methods) – Project Proposal Form

Instructions:

- Complete all sections of the form as thoroughly as possible.
- Ensure clarity, relevance, and technical alignment with your discipline.
- Submit this proposal form for academic approval before commencing your dissertation/project.

1. Student and Programme Details

Student Name	Devi Bhargavi Neppalli
Student ID Number	250387731
Programme Title	MSc Data Science and Artificial Intelligence
Supervisor	Matthew Bickley
Allocated Topic	Stochastic Modelling of Infectious Diseases
Date of Submission	23/03/2026

2. Title of Research Project

Dissertation Project Title Propose a concise, descriptive, and discipline-specific title for your dissertation project (max 20 words):
Dissertation Project Title: A Stochastic Modelling Framework for Predicting the Spatiotemporal Dynamics of Infectious Disease Outbreaks.

3. Research Problem, Aim & Objectives

Research Problem & Rationale Describe the research problem clearly and concisely (max 200 words)
Research Problem & Rationale: A significant limitation in current modeling is the inherent variability of human interaction. Most traditional models used by health officials are 'deterministic' (Kermack & McKendrick, 1927), utilizing fixed parameters that assume uniform interaction patterns. However, human transmission is influenced by demographic stochasticity, where random individual movements can drastically alter outbreak trajectories, particularly in high-density urban

environments. Currently, a research gap exists because many models operate at a national scale and overlook localized, random 'super-spreader' events (Lloyd-Smith et al., 2005) occurring in specific districts. If models do not account for this randomness, outbreak predictions will be inaccurate. By refining these models, we can assist city planners and clinicians in preparing for future health crises through precise spatiotemporal forecasting.

Research Aim

Provide the main aim(s) of your research and explain its relevance to your subject discipline or industry practice and its potential impact (e.g. technical, economic, environmental, academic) (max 200 words)

Research Aim:

The main aim of this research is to develop and test a computer-based mathematical model that specifically accounts for the random nature of disease transmission. Instead of just giving one "fixed" answer, this model will use probability to show a range of different possible outcomes for an outbreak. This aligns directly with my studies in Data Science and AI because it uses advanced computational techniques to solve a high-impact, real-world problem.

The potential impact of this work is significant. Technically, it provides a more robust tool for predicting disease peaks. Economically, it can help governments save money by targeting vaccinations and lockdowns only where they are truly needed, rather than shutting down entire regions. Academically, this project contributes to the field by showing how "noise" and randomness in data are not just errors, but essential pieces of information for accurate forecasting in modern epidemiology.

Research Objectives

Provide 3-5 specific, measurable objectives (max 100 words)

To Review: Analyze existing academic studies on how random mathematical patterns (stochastic processes) are currently applied to health data.

To Develop: Build a custom disease-tracking simulation using Python and data science libraries like NumPy or PyTorch.

To Simulate: Run multiple "what-if" scenarios to measure how random individual choices affect the overall speed of a virus spreading.

To Validate: Compare the predictions of the model against real-world historical data to ensure the results are accurate and reliable for future use.

4. Preliminary Literature Review

Preliminary Literature Review

Provide key findings from existing literature. Use and cite **at least 20 literature sources**, that should be **high quality peer reviewed journal articles**. (max 1200 words)

Introduction to Disease Modelling

The foundation of infectious disease modeling is rooted in deterministic frameworks. For over a century, the primary way scientists tracked diseases was through "Deterministic" models. The most famous is the SIR model, which stands for Susceptible, Infectious, and Recovered. This model assumes that if you know how many people are sick today, you can use a fixed formula to tell exactly how many sick tomorrows will be.

Early work by Karnack and McKendrick (1927) and Anderson and May (1991) created the foundation for this. However, as our data has become more precise, we have realized that these While deterministic models provide a foundational understanding of disease trends, they often fail to account for the stochastic variability inherent in individual transmission events, which can lead to significant discrepancies in localized forecasting.

The Shift to Randomness (Stochasticity)

The core of this research is the move toward Stochasticity. Stochastic modeling incorporates probabilistic variables to account for demographic uncertainty. Unlike deterministic models, stochastic models acknowledge that human

life is messy and unpredictable. Allen (2017) explains that "demographic stochasticity" the random chance of who you walk past at the supermarket or who sits next to you on a bus—is what determines if a small cluster of cases turns into a massive outbreak or just disappears.

Researchers like Bailey (1975) and Bartlett (1960) were the pioneers who argued that we should stop looking at diseases as a single predictable line and start looking at them as a "cloud" of possibilities. This is especially important for small groups, like a single school or a city office, where one random "unlucky" event can change everything.

Understanding "Super-Spreaders" and Local Clusters

One of the biggest problems with old models is that they treat everyone as if they have the same chance of spreading the virus. Lloyd-Smith et al. (2005) discovered that this is not true. Most people do not spread the virus at all, while a few "Super-Spreaders" spread it to dozens of people by random chance.

Stochastic modelling is the only way to capture these "outlier" events. Keeling and Rohani (2011) emphasize that "Spatial Dynamics"—the way people move through specific city streets—creates pockets of infection. By focusing on these random, local movements rather than national averages, we get a much clearer picture of how a virus behaves in a city environment.

The Role of Modern Computers and Algorithms

Because the math for "randomness" is incredibly hard for humans to calculate, we rely on specific computer algorithms. Gillespie (1977) created a famous method that allows a computer to simulate every single infection event one by one.

Today, we use Monte Carlo simulations. Monte Carlo methods facilitate the execution of the simulation across 10,000 iterations. By incorporating slight variations in random parameters during each iteration, the model identifies the most statistically probable outcomes for an outbreak. Monte Carlo methods facilitate the execution of the simulation across 10,000 iterations. By incorporating slight variations in random parameters during each iteration, the model identifies the most statistically probable outcomes for an outbreak. Each time, the computer changes the random events slightly. By looking at all 10,000 "games," we can see what is most likely to happen. Black and McKane (2012) show that these simulations are the most powerful tool we have for dealing with the uncertainty of modern pandemics.

The Future: AI and Machine Learning Synergy

Recent advancements in the field involve the integration of...combining these random models with Artificial Intelligence (AI). Researchers like King et al. (2015) and Wang et al. (2020) are using "Neural Networks" to help the models learn from real-time data. This is where Data Science and AI meet Epidemiology. Instead of humans guessing the "noise" in the data, the AI learns the patterns of randomness itself. This "Deep Learning" approach helps bridge the gap between old-school math and the high-tech needs of modern public health.

20 Literature Sources

Allen, L. J. S. (2017). A primer on stochastic epidemic models. (Explains the basic math of randomness).
Anderson, R. M., & May, R. M. (1991). Infectious Diseases of Humans. (The classic "Bible" of disease tracking).
Bailey, N. T. J. (1975). The Mathematical Theory of Infectious Diseases. (Early work on using probability).
Bartlett, M. S. (1960). Stochastic Population Models. (The foundation for random population math).
Bjørnstad, O. N. (2018). Epidemics: Models and Data using R. (How to use programming for disease data).
Black, A. J., & McKane, A. J. (2012). Stochastic formulation of ecological models. (Why is randomness essential).
Brauer, F., & Castillo-Chavez, C. (2012). Mathematical Models in Population Biology. (Covers the shift from fixed to random models).
Diekmann, O., et al. (2013). Mathematical Tools for Understanding Infectious Disease. (Focuses on modern

computer tools).

Gillespie, D. T. (1977). Exact stochastic simulation. (The invention of the simulation method used in Python).

Grassley, N. C., & Fraser, C. (2008). Mathematical models of infectious disease transmission. (A great overview for beginners).

Keeling, M. J., & Rohani, P. (2011). Modeling Infectious Diseases in Human and Animals. (Excellent resource for spatiotemporal dynamics).

Kermack, W. O., & McKendrick, A. G. (1927). A contribution to the mathematical theory of epidemics. (The original SIR model paper).

King, A. A., et al. (2015). Statistical Inference for Partial Models. (Combining math with real-world noisy data).

Kucharski, A. J., et al. (2020). Early dynamics of transmission of COVID-19. (Shows how these models worked during the recent pandemic).

Lloyd-Smith, J. O., et al. (2005). Superspreading and the effect of individual variation. (The most important paper on random super-spreaders).

Nasell, I. (2002). Stochastic models of some endemic infections. (Focuses on diseases that stay in a population for a long time).

Pastor-Satorras, R., et al. (2015). Epidemic processes in complex networks. (How diseases move through social networks).

Riley, S. (2003). Transmission dynamics of SARS in Hong Kong. (A real-world example of localized urban outbreaks).

Vespignani, A. (2012). Modelling dynamical processes in complex systems. (How to model a high-tech, connected world).

Wang, L., et al. (2020). Inference of stochastic epidemic models using deep learning. (The cutting-edge use of AI in this field).

Identify Research Gap

Identify gaps your research will address. Mention relevant theories, frameworks, or technologies (max 200 words)

Despite the vast amount of research on infectious diseases, there is a significant gap in how we handle localized uncertainty in urban environments. Most existing models are designed for national or global scales; they provide a "big picture" view but fail to account for the unique, random behaviors that happen at a neighborhood level. For instance, a single random event like a crowded local festival or a busy transit hub can cause a massive "super-spreader" spike that national models simply cannot predict.

Furthermore, there is a technical gap in computational efficiency. While "random" (stochastic) models are more accurate than fixed ones, they are often so mathematically heavy that they take too long to run. During an active outbreak, health officials need answers in hours, not days. There is a lack of accessible tools that balance this high-level accuracy with real-time speed.

Finally, while Artificial Intelligence is advancing rapidly, there is limited research on integrating Deep Learning (specifically LSTMs or Graph Neural Networks) to help refine the "random noise" in these disease models. My project aims to fill these gaps by exploring how modern AI can make localized, random models both faster and more precise for city-level planning.

Key Reference List

Out of your complete list of references (of at least 20, as above), list your **top 5 key** references here:

Key References:

1. Allen (2017) – Best for basic stochastic principles.
2. Keeling & Rohani (2011) – Essential for understanding city-level spread.
3. Gillespie (1977) – The foundation for the Python simulations you will write.
4. Lloyd-Smith et al. (2005) – Explains why "chance" events (super-spreaders) matter.
5. Wang et al. (2020) – Directly connects your project to the "AI" part of your degree.

5. Methodology and Data Strategy

Research Design and Methods

Explain your overall research design and approach (e.g. quantitative, qualitative, or mixed methods)
(max 300 words)

This research follows a quantitative design, focusing on the development and testing of computational simulation. Instead of using a traditional "deterministic" model—which produces the same result every time—I am designing a stochastic (random) framework. This is more realistic because it accounts for the fact that disease transmission is often based on chance encounters rather than fixed patterns.

The primary method involves building a customized version of the SIR (Susceptible-Infectious-Recovered) model using Python. This model will treat every individual interaction as a probabilistic event. For example, if a "Susceptible" person meets an "Infectious" person, the model does not automatically say they get sick; In this framework, each interaction is treated as a probabilistic event. The model applies a specific transmission threshold to determine the likelihood of an infection occurring, rather than assuming a fixed outcome.

To run these simulations, I will implement Gillespie's Algorithm (Gillespie's Algorithm) or Monte Carlo simulations. These methods allow the computer to run the same outbreak scenario thousands of times. By doing this, we can see a full range of possibilities from a small cluster that dies out quickly to a major epidemic. This "multi-run" approach is essential for identifying "worst-case" and "best-case" scenarios that simple models miss.

Finally, I will incorporate a spatiotemporal layer. This means the model will not just look at how many people get sick, but also where and when it happens in a localized city environment. I will compare the results of my random model against standard fixed models to prove that accounting for randomness provides a much more accurate tool for urban health planning.

Data Collection Strategy

Describe your intended data sources and collection/analysis techniques (max 200 words)

My strategy for this project is based entirely on secondary data. This means I will not be performing any clinical trials, interviewing patients, or collecting private medical records myself. Instead, I will gather high-quality, existing datasets that have already been made public by reputable health organizations.

The primary sources for my data will include the World Health Organization (WHO), the UK Health Security Agency, and academic repositories like Kaggle or GitHub that host historical outbreak data (such as COVID-19 or Influenza statistics). I am looking specifically for "anonymized" data information that shows infection trends, dates, and general locations without any names or personal IDs.

Once gathered, this data will be used to "calibrate" my model. For example, if historical data shows a virus typically spreads at a certain rate, I will use that rate as the baseline for my simulations. This ensures that even though my model is a simulation, it is grounded in real-world facts and behavior patterns observed in past outbreaks.

Resources Required (Tools, Techniques, Equipment/Technologies, Labs and workshops)

Specify tools, models, software, or technical frameworks you plan to use (max 100 words)

To complete this project, I require the following technical resources:

Hardware: A personal laptop with high-performance processing capabilities to handle intensive mathematical simulations.

Software: An Integrated Development Environment (IDE) like VS Code or Jupyter Notebooks.

Programming Language: Python, utilizing specialized data science libraries including NumPy and Pandas for data handling, and Matplotlib or Seaborn for creating visual charts.

AI Tools: PyTorch or TensorFlow will be used to integrate deep learning elements into the stochastic framework.

6. Ethical and Legal Considerations

Ethical and Legal Considerations Identify ethical, legal or professional issues relevant to your project (e.g. consent, data protection, AI use and bias, sustainability) (max 100 words)		
The main ethical priority for this project is protecting the privacy of the public while following data protection laws like GDPR. Since I am using health data to track disease trends, I must ensure that no individual can be identified from the numbers. Legally, I am required to use data that has been officially cleared for research use. Additionally, there is an ethical responsibility to ensure that the AI and mathematical models I build are fair and accurate, as biased results could lead to incorrect health advice that might negatively impact specific communities or neighborhoods.		
Proposed Mitigation Measures Describe how you will mitigate these issues (max 200 words)		
To manage these ethical and legal risks, my primary mitigation strategy is the strict use of fully anonymized secondary data. This means I will only download datasets from trusted organizations like the World Health Organization (WHO) or government health portals where all "identifying" information—such as names, specific house numbers, or personal ID codes—has already been removed. By using data that is already in the public domain for research purposes, I am ensuring that my project does not interfere with the private life of any individual. Furthermore, I will keep all research data stored securely on a password-protected laptop to prevent unauthorized access. From a technical standpoint, I will "double-check" my AI model by running validation tests against historical facts. This helps me identify any potential errors or biases in the code before the results are finalized. Finally, I will clearly document my data sources in the dissertation to maintain transparency. Because this project uses only anonymous public records and does not involve talking to human participants, the risk to individuals is extremely low, making it safe and ethically sound for academic study.		
Ethical Approval Do you need to apply for Research Ethics Approval? (Please discuss this in detail with your allocated dissertation supervisor) <i>Tick as appropriate on this form, then follow the instructions below.</i>	YES, Required ☐ My research <u>will</u> involve human participants or human subject data. <i>(This includes both:)</i> Primary data collection – e.g. interviews, surveys, focus groups, observations, physiological or biomedical data Secondary data use – e.g. using existing datasets that contain human data, even if you did not collect the data yourself (this includes online datasets, health records, organisational data, or purchased datasets)	NO, Not Required ☒ My research <u>will NOT</u> involve human participants or human subject data. <i>(This includes both:)</i> Primary data collection – e.g. interviews, surveys, focus groups, observations, physiological or biomedical data Secondary data use – e.g. using existing datasets that contain human data, even if you did not collect the data yourself (this includes online datasets, health records, organisational data, or purchased datasets)
What to do next You <u>must</u> complete an online Research Ethics	If your research project <u>does require</u> Ethics Approval you <u>must</u> complete the following online form:	If your research project <u>does not require</u> Ethics Approval you <u>must</u> complete the following online form:

Statement form. Please use the respective links provided in each column to do so.	• EP4DIS_Research Ethics Statement & Application	• EP4DIS_Research Ethics Statement
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7. Project Plan – Feasibility, Risks and Timeline

Timeline Overview Provide a realistic timeline using Gantt-style phases or week-by-week plan for the duration of the project.											
Timeline Overview: This project is designed to be completed over a 12-week period, divided into three main phases: Weeks 1–4 (Preparation Phase): The first month focuses on gathering all the necessary tools. I will spend this time reading existing research papers to build a strong theoretical foundation. I will also identify and download the public datasets (like COVID-19 or flu records) needed to test the model. Weeks 5–8 (Development Phase): This is the core technical stage. I will write the Python code for the stochastic model, setting up the "random" rules for how the virtual virus spreads. I will also begin building the AI/Deep Learning components to help refine the accuracy of the model. Weeks 9–12 (Analysis & Writing Phase): In the final month, I will run the simulations thousands of times to see the results. I will compare these "random" outcomes against traditional models to show improvement. The final weeks will be dedicated to writing the dissertation and preparing the final report for submission.											
Key Milestones Outline the key milestones of the project Milestone 1 (Week 4): Completion of the Literature Review and final selection of the historical datasets. Milestone 2 (Week 6): Successful build of the basic SIR Python model without errors. Milestone 3 (Week 8): Integration of Deep Learning elements (AI) into the stochastic framework. Milestone 4 (Week 10): Completion of all data simulations and comparative analysis. Milestone 5 (Week 12): Submission of the final Dissertation project.											
Risk Assessment and Mitigation Identify potential risks (e.g. access to data, equipment, delays) and how you will manage them (max 100 words) <table border="1"> <thead> <tr> <th>Risk</th><th>Impact</th><th>Mitigation</th></tr> </thead> <tbody> <tr> <td>The computer simulation might be too heavy for a normal laptop to run quickly.</td><td>It could delay my results.</td><td>I will start with a simpler version of the model first and use "clean" coding to make sure it runs as fast as possible.</td></tr> <tr> <td>Public health datasets might be incomplete or contain "noisy" data.</td><td>This could lead to inaccurate predictions or errors in the simulation.</td><td>I will use data only from trusted sources like the WHO and perform thorough data cleaning before starting.</td></tr> </tbody> </table>			Risk	Impact	Mitigation	The computer simulation might be too heavy for a normal laptop to run quickly.	It could delay my results.	I will start with a simpler version of the model first and use "clean" coding to make sure it runs as fast as possible.	Public health datasets might be incomplete or contain "noisy" data.	This could lead to inaccurate predictions or errors in the simulation.	I will use data only from trusted sources like the WHO and perform thorough data cleaning before starting.
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Complex stochastic algorithms can take longer than expected to debug.	This may cut into the time needed for final analysis and writing.	I will follow my 12-week schedule strictly and use pre-built Python libraries like NumPy to save development time.
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8. Approval Signatures

	Signatures	Date
Student	Devi Bhargavi Neppalli	20/03/2026
Supervisor		20/03/2026
Department Dissertation Lead		Click or tap to enter a date.
Module Leader (EPS)		Click or tap to enter a date.